

1) Solve the equation $x^{1/2}=12$

Note: Reciprocal of $\frac{1}{2} = 2$

a) $(x^{1/2})^2 = \underline{\hspace{2cm}}?$

b) Raise both sides of the equation $x^{1/2}=12$ to the power of 2.

$$(x^{1/2})^2 = (12)^2$$

$$x = \underline{\hspace{2cm}}?$$

2) Solve the equation $(x-3)^{3/2} = 8$

Note: Reciprocal of $\frac{3}{2} = \frac{2}{3}$ and $(x-3)^{3/2} = \sqrt[2]{(x-3)^3}$

a) $\left[(x-3)^{3/2}\right]^{2/3} = \underline{\hspace{2cm}} ?$

b) Raise both sides of the equation $(x-3)^{3/2} = 8$ to the power of $\frac{2}{3}$.

$$\left[(x-3)^{3/2}\right]^{2/3} = 8^{2/3} \quad \text{Note: IT-84 input: } 8^{2/3} = 8^{(2/3)}$$

$$x = \underline{\hspace{2cm}} ?$$

3) Solve the equation $(x-4)^{2/3} = 25$

Note: Reciprocal of $\frac{2}{3} = \frac{3}{2}$ and $(x-4)^{2/3} = \sqrt[3]{(x-4)^2}$

a) $\left[(x-4)^{2/3}\right]^{3/2} = \underline{\hspace{2cm}} ?$

b) Raise both sides of the equation $(x-4)^{2/3} = 25$ to the power of $\frac{3}{2}$.

$$\left[(x-4)^{2/3}\right]^{3/2} = \pm 25^{3/2} \quad \text{Note: IT-84 input: } 25^{3/2} = 25^{(3/2)}$$

Note: The original rational exponent is $\frac{2}{3}$. The numerator of $\frac{2}{3}$ is an even number.

Therefore we have to use " $\pm$ ".

$$x = \underline{\hspace{2cm}} ? \quad \text{and} \quad x = \underline{\hspace{2cm}} ?$$

4) Solve the equation $6x^{5/2} - 32 = 0$

Hint: Add 32 to both sides: $6x^{5/2} - 32 = 0 \Rightarrow 6x^{5/2} = 32$

Divide both sides by 6: $6x^{5/2} = 32 \Rightarrow x^{5/2} = 32/6 = 16/3$

Note: Reciprocal of $\frac{5}{2} = \frac{2}{5}$ and $x^{5/2} = \sqrt[2]{x^5}$

a) $[x^{5/2}]^{2/5} = \underline{\hspace{2cm}} ?$

b) Raise both sides of the equation $x^{5/2} = 16/3$ to the power of $\frac{2}{5}$.

$$[x^{5/2}]^{2/5} = (16/3)^{2/5} \quad \text{Note: IT-84 input: } (16/3)^{2/5} = (16/3)^{(2/5)}$$

Note: The original rational exponent is $\frac{5}{2}$. The numerator of $\frac{5}{2}$ is an odd number.

Therefore we do not use " $\pm$ ".

$$x = \underline{\hspace{2cm}} ?$$

